A QUASI-COMPACT MORPHISM OF ALGEBRAIC SPACES SATISFYING THE EXISTENCE PART OF THE VALUATIVE CRITERION IS UNIVERSALLY CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own valuative argument on points; the closedness conclusion is obtained by specialization, not by a citation of the scheme case.

Editorial note. The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Universally closed morphisms are detected on affine schemes mapping into the base. Lemma lemma-universally-closed-local, used at proof lines 8813-8815. Reduces to showing that |T x_Y X| -> |T| is closed for affine T over S.
- Stability of quasi-compactness and of the valuative criterion under base change. Lemmas lemma-base-change-quasi-compact and lemma-base-change-valuative-criteria, used at proof lines 8810-8813. Lets the hypotheses be assumed for the base changed morphism.
- Closedness of a subset of an affine scheme stable under specialization. Algebra, Lemma algebra-lemma-image-stable-specialization-closed, used at proof lines 8825-8828. Reduces the closedness of f(T) to stability under specialization.
- Existence of a valuation ring dominating a local ring inside a field. Algebra, Lemma algebra-lemma-dominate, used at proof lines 8844-8847. Produces the valuation ring A in K dominating the image of O_{Y,y}.
- An affine etale cover of a quasi-compact algebraic space. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover, used at proof lines 8819-8823. Turns the closed subset T into the preimage of a closed subscheme of an affine scheme.

Source reference. Original lemma 00IA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 672–672. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 660–660. Navigation pointer into the Stacks Project;

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the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4565–4565. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03IY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4620–4620. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03IT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4615–4615. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 01J5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2002–2002. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4555–4555. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *f is quasi-compact, and*
(2) *f satisfies the existence part of the valuative criterion.*

Then f is universally closed.

Proof. By Lemmas 03HF and 03IY properties (1) and (2) are preserved under any base change. By Lemma 03IT we only have to show that $|T \times_Y X| \to |T|$ is closed, whenever T is an affine scheme over S mapping into Y. Hence it suffices to prove: If Y is an affine scheme, $f : X \to Y$ is quasi-compact and satisfies the existence part of the valuative criterion, then $f : |X| \to |Y|$ is closed. In this situation X is a quasi-compact algebraic space. By Properties of Spaces, Lemma 03H6 there exists an affine scheme U and a surjective étale morphism $\varphi : U \to X$. Let $T \subset |X|$ closed. The inverse image $\varphi^{-1}(T) \subset U$ is closed, and hence is the set of points of an affine closed subscheme $Z \subset U$. Thus, by Algebra, Lemma 00HY we see that $f(T) = f(\varphi(|Z|)) \subset |Y|$ is closed if it is closed under specialization.

Let $y' \rightsquigarrow y$ be a specialization in Y with $y' \in f(T)$. Choose a point $x' \in T \subset |X|$ mapping to y' under f. We may represent x' by a morphism $\mathrm{Spec}(K) \to X$ for some field K. Thus we have the following diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x'} & X \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(\mathcal{O}_{Y,y}) & \longrightarrow & Y, \end{array}$$

see Schemes, Section 01J5 for the existence of the left vertical map. Choose a valuation ring $A \subset K$ dominating the image of the ring map $\mathcal{O}_{Y,y} \to K$ (this is possible since the image is a local ring and not a field as $y' \neq y$, see Algebra, Lemma

00IA). By assumption there exists a field extension K'/K and a valuation ring $A' \subset K'$ dominating A, and a morphism $\mathrm{Spec}(A') \to X$ fitting into the commutative diagram. Since A' dominates A, and A dominates $\mathcal{O}_{Y,y}$ we see that the closed point of $\mathrm{Spec}(A')$ maps to a point $x \in X$ with $f(x) = y$ which is a specialization of x'. Hence $x \in T$ as T is closed, and hence $y \in f(T)$ as desired. $\square$